

Omnitrix - Sovereign Second Brain

MSRIT Hackathon, Track 1 (Sovereign AI). Agents, tools, technology and features.

The idea: a private second brain with a team of AI agents that works like a personal secretary for a busy businessman. It understands your work, remembers everything, builds your to-do list and plans your day, re-plans when you miss something, keeps your promises, checks facts, acts through tools, deals with the outside world safely, and reaches you on your smartwatch. Everything runs on your own laptop - no data goes to outside AI companies.

Type	Agents	Building for the hackathon
Internal - core	13	13
Internal - extra	3	0
External	6	2 (Diplomat, Scout)
Total	22	15

Model key used in this document: **14B** = Qwen3 14B (thinking), **4B** = Qwen3 4B (fast), **Code** = mostly plain code, AI only where needed. All models run locally through Ollama.

Internal agents - core (13)

These work inside the company and take part in every problem.

#	Agent	What it does	Model	Build
1	Chief of Staff	The leader: turns every request into a plan and assigns steps to the other agents.	14B	Must
2	Librarian	The organizer: reads new emails, files, notes and voice notes, and pulls out tasks, people, promises and decisions.	4B	Must
3	Researcher	The finder: answers questions with proof, finds who knows what, prepares meeting briefs.	14B	Must
4	Historian	The diary: remembers past decisions and how things changed, and warns if we contradict them.	4B	Must
5	Promise Keeper	The reminder friend: tracks every promise made by you and to you, and makes sure it gets done.	4B + Code	Must
6	Planner (new)	The scheduler: builds your to-do list and day plan, and re-plans when something is missed.	Code + 4B	Must
7	Auditor	The checker: compares any checklist or question list against your documents and finds gaps.	4B + 14B	Should
8	Analyst	The "what if" thinker: when something changes, finds everything affected and the risks.	14B	Should
9	Writer	The drafter: writes emails, replies and proposals in your style, using only verified facts.	14B	Should
10	Fact Checker	The double-checker: catches wrong numbers, dates and contradictions before anything goes out.	4B + Code	Must
11	Operator	The doer: sends emails, books meetings, creates tasks - the only agent that acts.	4B	Must
12	Guardian	The security guard: checks permissions, blocks tricks, asks you to approve risky actions, logs everything.	Code	Must

#	Agent	What it does	Model	Build
13	Herald	The messenger: decides what matters and sends it to your phone and Galaxy Watch at the right time.	Code	Must

Internal agents - extra (3)

#	Agent	What it does	Model	Build
14	Interviewer	Finds knowledge only one person has and interviews them before it is lost.	14B	Later
15	Toolsmith	Builds a new tool when the agents need one that does not exist yet.	Coder model	Later
16	Apprentice	Notices tasks you repeat and offers to automate them.	4B	Later

External agents (6)

These deal with the outside world. Everything they send or receive passes the Guardian at the gate.

#	Agent	What it does	Model	Build
17	Diplomat	Talks to other people's and companies' AI agents for us, sharing only what we allow.	14B	Must
18	Scout	Watches the outside world (news, government rules, competitors) and reports what matters.	4B	Must
19	Client Desk	Answers client questions using only public or approved information.	4B	Later
20	Vendor Agent	Tracks supplier bills and renewals and spots unusual charges.	4B	Later
21	Filing Agent	Fills government and official forms, then stops for a human to submit.	4B	Later
22	Guest Agent Manager	Hires an outside expert AI for one job, shares only the needed piece, then disconnects.	Code	Later

Technology

Organizers' suggested technologies - our choice

Technology	Verdict	Role in our system
Ollama	Use - core	Runs the model for every agent: 14B for thinking, 4B for fast jobs, plus the embedding model. Forces answers into a fixed JSON format.
PostgreSQL + pgvector	Use - core	Stores everything: documents, search vectors, knowledge graph, tasks, schedule, promises, decisions, agent board, events, approvals, audit log.
Mem0	Use - core	Remembers your habits and preferences, plus each agent's private memory. Runs on Ollama + pgvector, so it stays local.
Obsidian	Use - core	Your window into the brain: daily plan, to-do list, decisions and people pages as Markdown files you own. Ticking a task marks it done.
Graphiti	Stretch	Upgrade for Historian and Analyst (facts that change over time). Heavy on local models - start with Postgres tables.
GBrain	Alternative	Ready-made people / company / meeting brain, but TypeScript/Bun and no scheduling - would add a second language.
llama.cpp	Underneath	Already used through Ollama.
vLLM	Later	For a future company server on NVIDIA GPUs. Not suited to a Mac laptop.
Letta	Skip	A full agent framework - would compete with our own agent system.
Zep	Skip	Mainly a cloud service - goes against sovereignty. Its open-source core is Graphiti.
Utopia	Skip	Enterprise-scale world model - too heavy for a personal secretary.

Other tools we need

Tool	Why	Used by
MCP Python SDK (FastMCP)	Build all our tool servers	All agents
ntfy (self-hosted)	Notifications to phone and Galaxy Watch, no cloud	Herald
Mailpit	Local email inbox and sending, no internet needed	Operator, Librarian, Client Desk
whisper.cpp	Voice notes to text, locally	Librarian
Tesseract (OCR)	Read scanned documents and photos	Librarian
APScheduler	Timers: morning brief, hourly checks	Promise Keeper, Herald, Scout, Planner
Postgres LISTEN/NOTIFY	Event bus - agents wake each other up	All agents
FastAPI	Dashboard and approval buttons	Guardian, Herald, user
Playwright MCP	Browser automation	Scout, Filing Agent
OR-Tools (optional)	Smarter scheduling	Planner

Architecture

Layer	What is in it
You	Dashboard (FastAPI), Obsidian vault, Galaxy Watch
Agents	15 agents for the hackathon, each with its own role, triggers, model, allowed tools and autonomy level
Memory	Mem0 (habits + agent memory); knowledge graph in Postgres tables (Graphiti later)
Gate	Guardian proxy - every tool call is checked, logged and, if risky, sent for approval
Tools	12 MCP servers built with the Python MCP SDK
Storage	PostgreSQL + pgvector - single source of truth
Models	Ollama on the laptop: Qwen3 14B, Qwen3 4B, embedding model

Agents wake up in three ways: when you ask (Chief of Staff plans it), when something happens (event bus), or on a timer (scheduler). AI is used only for understanding and writing; ranking, scheduling, missed-task checks and security rules are plain code.

Agents, technology and tools

What powers each hackathon agent (15)

Agent	Model	Memory	MCP servers	Extra tech
Chief of Staff	14B	Mem0 (your goals)	none - delegates	Mission runner from society-of-agents
Librarian	4B	-	files, knowledge, email	whisper.cpp, Tesseract, pgvector
Researcher	14B	-	knowledge	pgvector + keyword search
Historian	4B	Postgres timeline (Graphiti later)	memory, knowledge	-
Promise Keeper	4B + Code	Mem0 (who is usually late)	memory, tasks, knowledge	APScheduler
Planner	Code + 4B	Mem0 (your habits)	tasks, calendar, obsidian, prefs	Our own scheduling algorithm
Auditor	4B + 14B	-	knowledge	-
Analyst	14B	-	knowledge, memory	Graph queries in Postgres
Writer	14B	Mem0 (your writing style)	knowledge, prefs	-
Fact Checker	4B + Code	-	knowledge, memory	-
Operator	4B	-	email, calendar, tasks, obsidian	Mailpit
Guardian	Code	-	guardian, notify	Gates from society-of-agents
Herald	Code	Mem0 (what you ignore)	notify, calendar	ntfy, Galaxy Watch
Diplomat	14B	-	agent-link	Second copy of the system as the other side
Scout	4B	-	web	Local "outside world" folder for the demo

The 12 MCP tool servers

Server	Tools	Backed by	Used by
files	read_file, watch_folder, update_file, scan_image	Local disk + Tesseract	Librarian, Operator
knowledge	search_meaning, search_keywords, query_graph, add_to_graph, find_expert	Postgres + pgvector	Librarian, Researcher, Analyst, Auditor, Fact Checker, Writer
memory	log_decision, why_decided, get_timeline, track_promise, check_evidence	Postgres (Graphiti later)	Historian, Promise Keeper
prefs	remember_habit, get_preferences	Mem0	Planner, Herald, Writer, Chief of Staff
tasks	create_task, list_tasks, mark_done, mark_missed	Postgres	Planner, Promise Keeper, Operator
calendar	list_events, create_event, check_if_busy, find_free_slots, hold_slot, release_hold	Postgres (.ics import)	Planner, Operator, Herald
obsidian	write_daily_plan, read_ticks, write_note	Obsidian vault (Markdown)	Planner, Operator, Historian
email	read_email, send_email, reply_in_thread	Mailpit	Librarian, Operator, Client Desk
notify	send_alert, send_approval_button	ntfy to watch	Herald only
guardian	check_permission, check_rules, detect_trick, ask_approval, log, undo, redact	Postgres audit tables	Every tool call passes through it
web	web_search, read_webpage	Playwright / local folder	Scout only
agent-link	send_message, receive_message	HTTP between two instances	Diplomat, Guest Manager

Key features

Feature 1 - To-do list and day planning

The second brain reads your work, builds a to-do list, plans your day, and re-plans automatically when you miss something.

Step	What happens	Agents
1. Collect	Tasks come from emails, meetings, promises, voice notes and your requests.	Librarian
2. Understand	What it is, how long it takes you, how important, deadline, what it is waiting on.	Librarian, Researcher
3. Rank	Scoring formula: deadline, importance of the person, money involved.	Planner (code)
4. Fit into the day	Hard work in your focus time, travel time, breaks, protect gym and family time.	Planner (code) + Mem0 habits
5. Ask you	Plan sent to your watch: [OK] [Change].	Herald, Guardian
6. Missed? Re-plan	Time passed and not done - move it, bump lower-priority tasks, or ask you.	Promise Keeper, Planner

Feature 2 - Smart meeting replies

An email asks you to join a 5 PM meeting. The agents check your calendar; if you are free they draft a reply, you approve on your watch, and the reply is sent and the meeting is added to your calendar.

Main flow - you are free

#	Agent	What it does
1	Librarian	Recognizes a meeting request: who (Mehta), when (today 5:00 PM), length, topic, meeting link.
2	Researcher	Checks how important the sender is: key client, deal in progress.
3	Planner	Calendar shows 5:00-5:30 is free and your habits have no conflict. Holds the slot while waiting.
4	Writer	Drafts the reply in your style: "Hi Mehta, 5 PM works for me. See you on the call."
5	Fact Checker	Makes sure the time, date and name in the reply are correct.
6	Guardian	External email in your name - approval required.
7	Herald	Watch buzzes: "Mehta - Today 5:00 PM - You're free. [Accept] [Decline] [Suggest other time]"
8	You	Tap Accept on the Galaxy Watch.
9	Operator	Sends the reply in the same email thread, adds the meeting to the calendar, updates the Obsidian plan.
10	Promise Keeper	Tracks "I'll join at 5" and reminds you at 4:50. Guardian logs everything.

Smarter cases

Situation	What the agents do
You are busy at 5	Finds free slots and drafts: "I'm tied up at 5 - could we do 6 PM today or 11 AM tomorrow?"
Free, but it is your focus time	"Accept and move the proposal to 6 PM?" - Planner re-plans after you agree.
Clashes with something less important	Offers to accept and move the low-priority item.
Clashes with personal time	Never offers gym or family time; suggests other slots.
Very important sender	Marked urgent so the watch buzzes right away.
Time is unclear	Asks you, or offers 2-3 slots in the reply.
Different time zone	Converts it: "5 PM London is 9:30 PM for you."
Suspicious email	Guardian flags it; no automatic reply.
You do not respond	Reminds you again; never sends without your OK; request expires after the time passes.
You want changes	Voice on the watch: "Accept, but say I'll join 10 minutes late."

Rules

Rule	Why
Never reply without your approval	It is an external email in your name.
Hold the slot while waiting	Nothing else gets booked at that time in the meantime.
Protect personal time	Family time and gym are never offered.
Log everything	You can always see what was sent and why.
Learn from your choices	If you always decline Friday evenings, stop offering them.

New tools for this feature: `calendar.hold_slot`, `calendar.release_hold`, `email.reply_in_thread`.

Feature 3 - Smartwatch notifications (Samsung Galaxy Watch)

When something needs a human, the Herald taps you on the wrist. You tap Approve or Reject on the watch and the agents continue. Everything stays on our own devices.

How it flows

Step	What happens
1. Agents	Work on the laptop and reach something that needs a human.
2. Guardian	Marks the action as needing approval.
3. Herald	Decides it is urgent and sends it to our own ntfy server on the laptop.
4. Phone	The ntfy app (same Wi-Fi / hotspot) receives it.
5. Galaxy Watch	Buzzes and shows the message with Approve / Reject buttons.
6. Tap	Goes back to the laptop, gets logged, and the Operator carries out the action.

What shows on the watch

Type	Example
Urgent (buzz now)	Approvals ("Mehta - 5 PM - You're free [Accept]"), security alerts.
Important (good time)	Overdue promises, missed tasks and the new plan, contradictions with past decisions.
Daily summary	Morning brief (today's plan, promises due) and evening wrap-up.
Voice reply	"Approve, but change 24 hours to 48 hours."

Safety rules

Action	Watch tap enough?
Internal reminder, book meeting	Yes
Email to a client / meeting reply	Yes, with short preview
Sign a contract / pay money	No - phone + fingerprint needed
Delete data	No - laptop only

Keeping it sovereign

- Own notification server (ntfy) on the laptop - no Google or Samsung cloud in the path.
- One-time approval tokens, so nobody else on the Wi-Fi can approve actions.
- Use our own phone hotspot with mobile data off - proves nothing needs the internet.
- Samsung phone: set ntfy battery to Unrestricted, or notifications stop arriving.
- Backup: if the watch fails on stage, the same notification is on the phone.

Hackathon build plan

Phase	Build	Checkpoint
0. Setup	Ollama models, Postgres, shared Task format, fake demo data, test set	Everyone can run the project
1. Understand	Librarian extracts tasks, people, promises	Drop in emails - to-do list appears
2. Plan	Ranking + Planner fits tasks into the calendar	To-do list becomes today's schedule
3. Re-plan	Missed-task detection + re-plan + fast-forward demo clock	Jump the clock - new plan
4. Watch + meetings	Herald, Guardian, ntfy, smart meeting replies	Email arrives - watch buzzes - reply sent
5. More agents	Historian, Researcher, Writer, Fact Checker, Auditor, Analyst	Richer answers, errors caught
5.5 Fine-tune (stretch)	Only if the base 4B scores below 90% on the test set: LoRA fine-tune the Librarian's extraction model on the laptop	Before/after accuracy, e.g. 78% to 93%
6. Outside	Diplomat + Scout	Two secretaries agree on a meeting time
7. Finish	README, prior-work disclosure, diagram, rehearsal, backup video	Ready to submit

Phase 5.5 - Fine-tuning the Librarian (stretch goal)

Fine-tuning is not required by the brief - the judges want a working system first. Knowledge (people, deals, facts) always comes from memory and the database, never from fine-tuning. Fine-tuning only teaches the small model to do one job better: pulling tasks and meeting requests out of text in our exact format.

Details	
When	Only after Phase 4 works. First measure the base Qwen3 4B on our 30-item test set: 95% or more - skip it; below 90% - do it.
What	The Librarian's extraction model (Task Candidate and Meeting Request formats). Optional later: the Writer, on your own emails, to match your style.
Training data	Teacher to student, all local: the 14B model labels examples, the 4B model learns from them; we hand-check a sample.
How	LoRA fine-tuning with MLX-LM (Apple's framework) directly on the M5 Pro; then load the model locally (Ollama after conversion, or the MLX server).
Time	About 3-4 h for data, about 1 h to train, 1-2 h to evaluate and plug in.
Don't fine-tune	The Planner (plain code), the Guardian (rules), or anything that holds facts.
Pitch line	"We fine-tuned our own model on our own laptop - no data, no training and no model ever left the machine."

Reuse from society-of-agents (disclose as prior work): Ollama provider, agent board, gates, audit log, dashboard base. Cloud providers are removed from the agent path.

"A secretary that knows everything about your work, and tells no one."